

VPC – Virtual Private Cloud

❖ What is VPC ?

- A **VPC** (Virtual Private Cloud) is a **private, isolated section of a public cloud** where you can run your resources (such as virtual machines, databases, and storage) in a **secure** and controlled environment.
- It is commonly used in cloud platforms like **AWS (Amazon Web Services)**, **Google Cloud Platform (GCP)**, and **Microsoft Azure**.
 - Managed in the cloud. With a VPC, you can control network settings such as:
 1. IP address ranges
 1. Subnetting
 2. Routing
 3. Internet access
 4. Firewall rules (Security Groups & Network ACLs)

✓ Why Use a VPC

1. Network Isolation
2. Enhanced Security
3. Custom Network Configuration
4. Scalability and Flexibility
5. Internet and On-Premises Connectivity
6. High Availability & Disaster Recovery
7. Cost Management

✓ Key Components of a VPC

Component	Description
CIDR Block	IP range used by the VPC (e.g., 10.0.0.0/16)
Subnets	Sub-divisions of the VPC IP range. Can be public or private
Internet Gateway (IGW)	Enables communication between instances in public subnet and the internet
NAT Gateway / NAT Instance	Allows instances in private subnets to access the internet without exposing them to incoming connections
Route Table	Controls routing of traffic within the VPC and to external networks
Security Groups	Virtual firewalls that control inbound and outbound traffic at the instance level
Network ACLs	Firewalls that control traffic at the subnet level

❖ What is a Subnet?

- A subnet is a **smaller network segment** within a VPC.
- Subnets **divide the IP range** of the VPC into smaller segments to isolate resources logically and functionally.

➤ Types of Subnets:

- **Public Subnet:** Connected to the internet via an Internet Gateway.
- **Private Subnet:** Not directly accessible from the internet. Typically used for backend services or databases

❖ Purpose of Subnets

- Subnets, or **subnetworks**, are subdivisions of an IP network that serve several important purposes in network design and management:
- Separate frontend (web servers) and backend (databases).
- Deploy resources across multiple Availability Zones for high availability.

❖ Steps to Create a VPC and Subnets

1. Create a VPC

▪ Create Subnets:-

1. Each subnet resides in a single Availability Zone.
2. Subnets are either public or private depending on routing and gateway configuration.

2. Add an Internet Gateway:-

- 1] To allow public access to resources in the public subnet.
- 2] Must be explicitly attached to the VPC.

3. Configure Route Tables:-

- 1] Public subnets use a route that points to the Internet Gateway.
- 2] Private subnets have no direct route to the internet (or use NAT for outbound-only access).

4. Assign Public IPs (if needed) :-

Resources in public subnets need public IPs or Elastic IPs to be accessible from the internet.

▪ Difference Between VPC and Subnet

Aspect	VPC	Subnet
Definition	A virtual private network in the AWS cloud	A subnet is a segment within a VPC, dividing the VPC's IP range
Scope	Represents the entire network environment	Represents a smaller network within the VPC
CIDR Block	Assigned a large CIDR block (e.g. /16)	Assigned a smaller CIDR block from the VPC range
Purpose	Provides a private, isolated environment to launch cloud resources (EC2, RDS, etc.).	Organizes and segments the VPC network into smaller, manageable blocks.
Internet Access	Requires IGW, route tables, etc	Can be public or private based on route table
Availability Zone	Can span multiple AZ	Exists in one specific AZ

[GENERAL INTERVIEW QUESTIONS]

Q1. What is a VPC?

- A VPC (Virtual Private Cloud) is an isolated network in the cloud where you can launch AWS resources in a custom IP range.

Q2. Why do we use CIDR in VPC?

- CIDR (e.g., 10.0.0.0/16) defines the IP address range for a VPC and helps in creating subnets by dividing this range.

Q3. What is a Route Table in VPC?

- It contains rules (routes) that decide how traffic moves within the VPC and outside.

Q4. Can we create multiple subnets in one VPC?

- **Yes** A VPC can have many subnets, usually spread across different availability zones.

Q5. How many IPs are reserved in each subnet by AWS?

- AWS reserves 5 IP addresses per subnet (network address, broadcast, and three AWS internal uses).

Q6. Explain how NAT Gateway works and where it should be placed?

- A NAT Gateway (Network Address Translation Gateway) **allows** instances in a private subnet to connect to the internet **without** exposing those instances to inbound traffic from the internet.

Q7. What is the difference between VPC Peering and Transit Gateway? When would you choose one over the other?

- **VPC Peering:** A one-to-one connection between two VPCs.
- **Transit Gateway:** A centralized hub that connects multiple VPCs and on-premises networks.
- Use **VPC Peering** for a few connections, and **Transit Gateway** for managing 10+ VPCs efficiently

Q8. How does subnetting affect availability and fault tolerance in a VPC?

- Subnetting across multiple Availability Zones (AZs) ensures that resources are distributed. If one AZ fails, instances in another AZ continue to work, improving fault tolerance and high availability

Q9. Describe the steps required to make a subnet public?

- To set up high availability (HA) for NAT in a VPC, you need to ensure that resources in private subnets across multiple Availability Zones (AZs) can access the internet even if one AZ goes down.
 - **High Availability Setup for NAT in a VPC:**
 1. Create a VPC with at least two AZs
 2. Deploy a NAT Gateway in *each* public subnet
 3. Update route tables for private subnets
 4. Add an Internet Gateway to the VPC
 - **What is the smallest and largest subnet mask?**
 - **Smallest:** /28 → 16 IPs (11 usable)
 - **Largest:** /16 → 65,536 IPs (65,531 usable)

Q10. How would you troubleshoot an EC2 instance in a public subnet that cannot access the internet?

- **Check following steps:-**
 1. Instance has public IP or Elastic IP
 2. Subnet is associated with route table that routes 0.0.0.0/0 to IGW
 3. Internet Gateway is attached to VPC
 4. Security Group allows outbound traffic
 5. NACL allows outbound and inbound traffic

Q11. Virtual Private Cloud (VPC) Configuration

Q12. What is an Internet Gateway?

- An **Internet Gateway (IGW)** is a **horizontally scaled, and highly available** AWS component that allows **communication** between resources in your **VPC** and **the internet**.
- When EC2 instances need to send or receive traffic to or from the internet, they do this via the Internet Gateway. It is one of the most critical components for enabling public connectivity in a cloud network.
- **Key Functions:-**

Function	Description
Outbound Internet Access	Allows EC2 instances with a public IP to send traffic to the internet
Inbound Internet Access	Enables responses to return and allows traffic from the internet to reach public IPs
NAT Translation	Performs 1:1 NAT for public IPv4 addresses on EC2 instances
Highly Available	IGW is managed by AWS and scales automatically across multiple AZs

- ✓ **Conditions Required for Internet Access via IGW:-**
 1. To allow instances inside a VPC to connect to the internet using an IGW, all of the following conditions must be true:
 2. The VPC must have an Internet Gateway attached.
 3. The subnet should be connected to a route table that has a route pointing to the IGW (using destination 0.0.0.0/0).
 4. The instance must have a public IP or an Elastic IP.
 5. Security group rules must allow inbound and outbound traffic on required ports (e.g., SSH - port 22, HTTP - port 80, RDP - port 3389).
 6. *If any of these conditions is missing, internet access will not work*

Q13. Public vs Private EC2 Instances

- EC2 instances can be classified as public or private, based on whether they can communicate with the internet directly.

■ Public EC2 Instances:-

✓ These are instances that are:

1. Placed in a subnet that has a route to the Internet Gateway
 2. Assigned a public IP address or Elastic IP
- **Purpose:** Public instances are used when the instance needs to be accessed directly from the internet.
(Ex. use for frontend)
 - **Security Tip:**
 1. While public instances are necessary for some use cases, they must be properly
 2. secured with strict security group rules, firewalls, and monitoring.

■ Private EC2 Instances

✓ Private instances are those:

1. Placed in a subnet that does not have a route to the IGW
 2. Do not have a public IP
 3. These instances cannot be accessed directly from the internet.
- **Purpose:** They are used for internal systems that don't need internet exposure. *(use for backend)*

Q14. Why is the Internet Gateway Important in All This?

- The IGW is the **only way to provide internet connectivity to EC2 instances** within a VPC.
- It forms the **backbone of any public network** architecture inside AWS.
- **Without IGW**, even instances with public IPs won't work.
 - **It is also a key part of configuring services like:**
 1. Bastion hosts
 2. Web-based applications
 3. NAT Gateways
 4. Elastic Load Balancers (internet-facing)

❖ NAT GATEWAY

- A **NAT Gateway** is a managed service in AWS that **allows EC2 instances in private subnets to access the internet** or other AWS services, without allowing anyone from the internet to access them.
- It works like a **translator** or middleman, helping private servers go out to the internet (like for software updates), but blocking any incoming internet traffic, keeping them.

■ Purpose

- Helps private instances connect to the internet.
- Keeps private instances secure and hidden from outsiders.
- Allows only outbound (outgoing) internet traffic, not inbound.
- Works with both IPv4 and IPv6.

Q14. How Traffic Flows through NAT Gateway

Private instance wants to access the internet (like to update software).

1. It sends the request to the NAT Gateway.
2. NAT Gateway changes the instance's private IP to a public IP (using Elastic IP).
3. Request goes to the internet.
4. Internet sends back the response.
5. NAT Gateway changes the public IP back to the private IP.
6. The response reaches the private instance.

Note:- So, the outside world never sees the private IP—only the NAT Gateway does the Talking

Q15. How do you make a subnet public in a VPC?

1. Attach an Internet Gateway to the VPC
2. Create or update the subnet's route table with a route: 0.0.0.0/0 → igw-xxxx
3. Enable auto-assign public IP on EC2 launch.

Q16. How do you ensure public EC2 instances are secure?

1. Restrict Security Group rules
2. Disable root login
3. Use key pair authentication
4. Restrict SSH to specific IPs (e.g., office IP)

Q17. What is VPC Peering in AWS?

- A private connection between two VPCs that allows traffic to route between them using private IP addresses.

Q18. Can you peer VPCs in different AWS accounts or regions?

- Yes. VPC Peering supports cross-account and inter-region connections

Q19. What are the limitations of VPC Peering?

1. No transitive routing
2. No overlapping CIDRs
3. Manual route and security configuration
4. Doesn't support security group referencing across VPCs